EMR-MR Sudan — Project Planning Document

Oct 6, 2026

Summary

This document is the working plan for building EMR-MR, the national emergency response and patient records system for Sudan. It explains how the team works, who does what, the order of work, every sprint with its tasks, and how we test, report and handle risks.

Order of work (agile, step by step):

1. **Sprint 0** — set up AWS, code projects and pipelines; research how users work.
2. **Sprints 1–4** — build the database and all backend APIs first, while the designers finish all web and mobile screens and test them with real users.
3. **Sprints 5–8** — build the four web portals and the mobile app on top of the finished backend.
4. **Sprints 9–12** — connect everything end to end, harden security and speed, test with the client, train users and go live in Khartoum.

There are **13 sprints** of 2 weeks each [confirm sprint length]. Every task is in ClickUp (folder **EMR-MR Sudan** in the Development space) with a user story, a description, acceptance criteria for QA, and what it depends on. This document lists the tasks; ClickUp holds the full detail and the live status.

1. Project overview

Goal: when someone in Sudan calls **999**, the right hospital sends the right ambulance as fast as possible, and the patient's record follows them from the call to discharge.

How a case works: the Sudan Health Authority call centre answers the call, the system sets the urgency (P1–P4) automatically, the agent drops a pin (the system shows a 300 m area), and sends a request to the nearest working hospital. The hospital accepts within 5 minutes and sends one of its own ambulances. The crew uses the mobile app, sends a summary to the hospital, and hands the patient over. First-time patients get a full new record.

Release 1 includes:

Part	Users
Core system (Spring Boot API, PostgreSQL on AWS)	All parts use it
Admin portal	System admins
Health Authority call centre portal	Call-centre agents (one or more per shift)
Hospital emergency portal	ER doctors and nurses
Hospital staff portal	Ward, records and lab staff
Mobile app (Android): driver mode and paramedic mode	Hospital ambulance crews

All screens work in **Arabic and English** as equal languages. Login uses a second code by **SMS (Zain)**.

Not in Release 1 (Release 2): full medical-record tabs, DHIS2 reporting (no access yet), national ID check, large-incident mode, police and civil defence handoff, iPhone app.

2. How we work

We use **Scrum**: short sprints, a working demo at the end of each one, and clear rules for when a task can start and when it is finished.

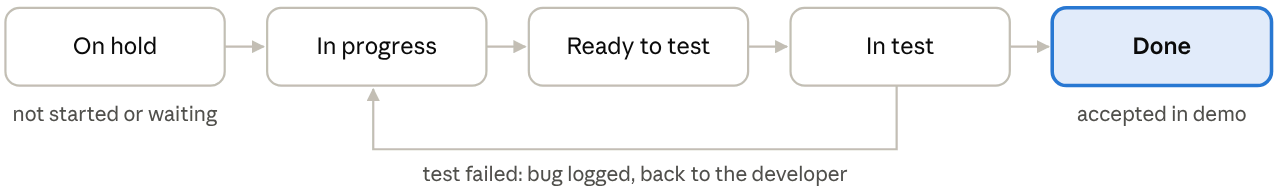
Sprint rhythm (every 2 weeks):

Meeting	When	Length	Purpose
Sprint planning	Day 1	2 hours	Choose tasks, confirm the sprint goal
Daily stand-up	Every day	15 minutes	Done yesterday, plan today, anything blocked
Refinement	Day 7	1 hour	Prepare and estimate the next sprint's tasks
Sprint review (demo)	Day 10	1 hour	Show working software; client and product owner accept

Meeting	When	Length	Purpose
Retrospective	Day 10	45 minutes	What to keep, change and try next

Task statuses in ClickUp:

Every task moves through five statuses; a failed test goes back



task statuses · 5 steps, 1 return path

Status	Meaning	Who moves it
On hold	Not started, or waiting for something	Project manager
In progress	A developer or designer is working on it	Developer
Ready to test	Code is on staging; QA can start	Developer
In test	QA is testing every acceptance criterion	QA
Done	Passed all tests and accepted in the demo	QA, then product owner

Definition of Ready (a task can start when):

- [ ] It has a user story, a description and acceptance criteria
- [ ] The design is approved, if it has a screen
- [ ] The API is agreed between backend and front end
- [ ] The tasks it depends on are done, or a stand-in exists
- [ ] It is estimated (story points)

Definition of Done (a task is finished when):

- [ ] Code is reviewed and merged
- [ ] Automatic tests pass

- [ ] QA passed every acceptance criterion on staging
- [ ] It works in Arabic and in English
- [ ] Role and scope checks pass
- [ ] Mobile tasks pass the network-off test
- [ ] Product owner accepts it in the demo

How every task is written in ClickUp:

Part	What it contains
Title	Track tag and short name, e.g. "[BE] Login with SMS code"
User Story	As a …, I want …, so that …
Description	Where it opens from, fields (required or not), numbered steps, sorting, pages, empty-state and error messages
Acceptance Criteria	Checks QA follows one by one, including errors
Affected by / Affects	Tasks it depends on, and tasks that depend on it
(Difficult part)	Anything hard, explained in brackets

Track tags: **[BE]** backend and database, **[UX]** design, **[WEB]** web portals, **[MOB]** mobile app, **[QA]** testing, **[OPS]** servers and AWS, **[PM]** project management.

3. Team and responsibilities

One Scrum team of 14–15 people from our company. Training and support after go-live are also done by our team (client decision).

Role	People	Responsible for
Project manager / Scrum master	1	Sprints, meetings, blockers, reports to the client, risks
Systems analyst / product owner	1	Requirements, task writing, priorities, accepting work in demos

Role	People	Responsible for
Tech lead / architect	1	Technical decisions, database design, code reviews, integrations
Backend developers (Spring Boot)	3	Database, APIs, priority function, dispatch, sync, integrations
Web developers (React)	2–3	Shared components and the four web portals
Mobile developers (Flutter)	2	Driver and paramedic app, offline sync, GPS
UI/UX designer	1	User research, design system, all web and mobile designs, usability tests
QA testers	2	Test plan, test cases, device lab, automatic and manual tests
DevOps engineer	1	AWS, pipelines, monitoring, backups, security settings

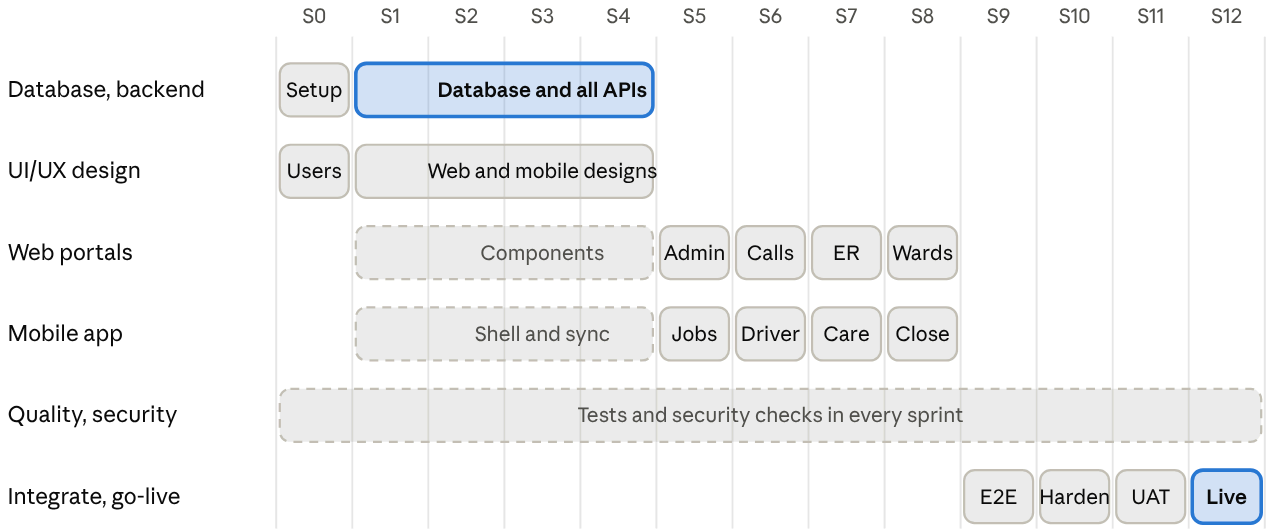
Who decides what:

Decision	Decides	Consulted	Informed
What is built and in which order	Product owner	Client, tech lead	Team
How it is built	Tech lead	Developers	Product owner
Whether a task is done	Product owner	QA	Team, client
Release to live	Project manager and client	Tech lead, QA	All users
Change of scope	Client and project manager	Product owner	Team

4. Roadmap

The backend and all designs are finished before screens are built, so the web and mobile teams build once on stable APIs. During Sprints 1–4 the web and mobile developers build shared components and app shells against API stand-ins (dashed bars).

Database and backend first; web and mobile build on finished APIs



roadmap · 13 sprints, 6 tracks

Horizon	Sprints	Focus	Confidence
Now	S0–S4	Database, backend APIs, all web and mobile designs	High: committed
Next	S5–S9	Four web portals, mobile app, full case end to end	Good: planned
Later	S10–S12	Hardening, client testing, training, go-live in Khartoum	Planned; dates set after S4
Release 2	After go-live	Full record tabs, DHIS2, national ID check, large incidents	Direction only

5. Sprint plan

Thirteen sprints, 79 tasks. Each sprint has a goal and a demo that proves the goal is met. Full task details (user story, description, acceptance criteria) are in ClickUp.

Sprint 0 — Setup

Goal: the team can build, test and deploy to staging on AWS; user research is done. **Demo:** a code change deploys to staging automatically.

Task	Priority	Affected by
[PM] Decide AWS main and backup regions	High	—
[OPS] AWS account, network and four environments	High	AWS regions
[OPS] CI/CD pipeline to staging	High	AWS environments
[BE] Spring Boot project setup and API skeleton	High	CI/CD pipeline
[WEB] React project setup with Arabic and English	Normal	CI/CD pipeline
[MOB] Flutter project setup with Arabic and English	Normal	CI/CD pipeline
[UX] User research: personas and key journeys	High	—
[QA] Test strategy and device lab plan	Normal	—
[PM] Backlog estimate, Definition of Ready and Done	High	—
[OPS] Monitoring and error tracking baseline	Normal	AWS environments

Sprint 1 — Database and login

Goal: full database design, login with SMS code, roles and scope, access records; design system and wireframes. **Demo:** a user logs in with an SMS code and every action is recorded.

Task	Priority	Affected by
[BE] Database design v1 (all core tables)	Urgent	Spring Boot setup
[BE] Login with SMS code (Zain), lockout and reset	High	Database; Zain SMS account
[BE] Roles and scope on every API call	High	Login
[BE] Access records (audit log)	High	Roles and scope
[BE] Users API (create, edit, disable)	Normal	Roles and scope

Task	Priority	Affected by
[UX] Design system (Arabic and English)	High	User research
[UX] Wireframes for all portals and the mobile app	High	User research
[WEB] Shared component library	Normal	Design system
[MOB] App shell, encrypted storage and login screen	Normal	Login API
[QA] Login, scope and access-record tests	High	Login; roles; access records

Sprint 2 — Master data APIs

Goal: areas (detailed Khartoum and all Sudan), landmarks, hospitals, hospital ambulances, lists and settings; final call centre and admin designs. **Demo:** Khartoum data loaded through the API.

Task	Priority	Affected by
[BE] Areas API: detailed Khartoum and all of Sudan	Urgent	Database; Khartoum area list from client
[BE] Landmarks API	Normal	Areas API
[BE] Hospitals API with status and beds	High	Roles and scope
[BE] Hospital ambulances API with QR codes	High	Hospitals API
[BE] Lists and settings APIs	Normal	Database
[UX] Final designs: call centre and admin	High	Wireframes
[WEB] Portal shells with language switch	Normal	Component library; login
[MOB] Sync client and send-later queue	Normal	App shell
[QA] Load and check Khartoum test data	Normal	Areas; hospitals; ambulances APIs

Sprint 3 — Cases and priority APIs

Goal: call intake, built-in priority function (P1–P4), pin with 300 m area, hospital request with 5-minute timer, live updates; hospital designs. **Demo:** a test case gets its priority automatically and reaches a hospital.

Task	Priority	Affected by
[BE] Call intake API (999)	High	Database
[BE] Built-in priority function (P1–P4)	Urgent	Call intake; triage model chosen
[BE] Location: pin with 300 m area and area lookup	High	Areas and landmarks APIs
[BE] Nearest working hospitals with a free ambulance	High	Hospitals; ambulances APIs
[BE] Hospital request with 5-minute timer	Urgent	Nearest hospitals
[BE] Live updates and priority-sorted queues	High	Priority function
[UX] Final designs: hospital ER and hospital staff	High	Wireframes
[WEB] Map and live-update components	Normal	Portal shells
[MOB] Job alert screen and offline queue	Normal	Sync client
[PM] Choose the standard triage model and P1 timer	High	—

Sprint 4 — Dispatch and patient APIs

Goal: hospital sends its own ambulance, job confirmation and phone-call log, GPS, mobile sync, patients and first-time record; final mobile designs and usability test. **Demo:** a full case runs through the backend; all designs approved.

Task	Priority	Affected by
[BE] Hospital sends own ambulance (job API)	Urgent	Hospital request; ambulances API

Task	Priority	Affected by
[BE] Job confirmation and phone-call log	High	Job API
[BE] GPS upload and positions API	High	Ambulances API
[BE] Mobile sync API with conflict rules	Urgent	Database
[BE] Patients API: search, ID types, merge	High	Database
[BE] First-time record, hospital checks, vital signs	High	Patients API
[UX] Final mobile designs (driver and paramedic)	High	Wireframes
[UX] Usability test with real users and design sign-off	High	All final designs
[WEB] Forms and tables components	Normal	Component library
[MOB] GPS service and offline maps base	Normal	App shell

Sprint 5 — Admin portal and app jobs

Goal: admin portal; app login, shift and job alert. **Demo:** admin sets up users and ambulances; a crew accepts a job.

Task	Priority	Affected by
[WEB] Admin portal	High	All master data APIs
[MOB] Login, PIN and shift start by QR	High	Ambulances API with QR codes
[MOB] Job alert, accept and confirmation	Urgent	Job API; job alert screen
[QA] Admin portal and app jobs tests	Normal	Tasks above

Sprint 6 — Call centre portal and driver mode

Goal: Health Authority call centre portal; driver mode. **Demo:** a 999 call becomes a request to the nearest hospital.

Task	Priority	Affected by
[WEB] Health Authority call centre portal	Urgent	Cases, priority, location and request APIs
[BE] Phone link: 999 caller number on screen	High	Call centre phone system (client)
[MOB] Driver mode	High	GPS service; job alert
[QA] Call centre and driver mode tests	Normal	Tasks above

Sprint 7 — Hospital ER portal and paramedic mode

Goal: hospital emergency portal; paramedic mode. **Demo:** the hospital accepts a request and sends its own ambulance.

Task	Priority	Affected by
[WEB] Hospital emergency portal	Urgent	Hospital request; job API; phone-call log
[MOB] Paramedic mode: patients, vital signs, danger score	High	Patients API; first-time record API
[QA] Hospital ER and paramedic tests	Normal	Tasks above

Sprint 8 — Hospital staff portal and handover

Goal: hospital staff portal with the full first-time record; summary, handover and close in the app. **Demo:** patient handed over; new record created.

Task	Priority	Affected by
[WEB] Hospital staff portal	High	Patients and first-time record APIs
[MOB] Summary to hospital, handover and close	High	Paramedic mode
[QA] Network-off and handover tests	High	Tasks above

Sprint 9 — End-to-end integration

Goal: a full case from call to discharge works across all parts; dashboard, reports, FHIR API. **Demo:** a full case runs live end to end.

Task	Priority	Affected by
[BE] End-to-end fixes, dashboard data and FHIR API	High	All portals and app
[WEB] Dashboard and exports	Normal	Dashboard data
[MOB] App working with all portals	Normal	All app modes
[QA] Every case path end to end	High	Tasks above

Sprint 10 — Hardening

Goal: the system is safe, fast and survives failures. **Demo:** failover, load and security tests pass.

Task	Priority	Affected by
[OPS] AWS failover, standby database, backups and restore	Urgent	AWS backup region
[QA] Load test at 3× the busiest expected time	High	Integration
[QA] Independent security test and fixes	Urgent	Integration
[MOB] Battery and speed tuning	Normal	App working with portals

Task	Priority	Affected by
[WEB] Speed fixes	Normal	Load test results

Sprint 11 — User acceptance testing

Goal: the Health Authority and hospitals test and accept the system. **Demo:** client sign-off.

Task	Priority	Affected by
[PM] User acceptance testing with the Health Authority and hospitals	Urgent	Hardening
[ALL] Bug fixing from acceptance testing	High	Test results
[PM] Training materials in Arabic and English	High	Final screens

Sprint 12 — Go-live

Goal: the system is live in Khartoum with close support. **Demo:** first real cases handled.

Task	Priority	Affected by
[OPS] Real data load: areas, hospitals, ambulances, users	High	Client data
[PM] Training sessions for call centre, hospitals and crews	High	Training materials
[MOB] Install the app on hospital phones	High	Phones bought by hospitals
[OPS] Go-live and close support with monitoring	Urgent	All above

Backlog — Release 2

Full medical-record tabs (medicines, labs, imaging, operations, vaccines, visits); DHIS2 export (when access is given); national ID check; large-incident mode; police and civil defence handoff; iPhone app.

6. Dependencies

These items come from outside the team. Each has an owner and a fallback so work does not stop.

Dependency	Needed by	Owner	If late
AWS account and approval for data outside Sudan	Sprint 0	Client and project manager	Start in a test AWS account; no real data until approved
Zain SMS account	Sprint 1	Client	Use a temporary SMS test account
Detailed Khartoum area list	Sprint 2	Client with our team	Start from OpenStreetMap; admins correct areas later
Standard triage model chosen	Sprint 3	Product owner	Build the priority function with settings; load the model when chosen
Phones models chosen by hospitals	Sprint 4	Hospitals	Test on the cheapest common Android phones
Real users for usability tests	Sprint 4	Health Authority and hospitals	Test with our staff first, then repeat with real users
Phone link for the 999 caller number	Sprint 6	Client and tech lead	Agents type the caller number until the link is ready
Hospitals ready for acceptance testing	Sprint 11	Health Authority	Test with a smaller group of hospitals first

7. Risks

The project manager reviews this list in every sprint planning meeting.

Risk	Impact	How we reduce it	Owner
Khartoum area data is incomplete (no street names)	Callers cannot be located quickly	Start early (Sprint 2); pin + 300 m area; landmarks; crews add landmarks	Product owner

Risk	Impact	How we reduce it	Owner
Zain SMS is slow or down	Nobody can log in	Monitor SMS delivery; alert support; temporary second SMS account [confirm]	DevOps
Hospitals answer requests slowly	Patients wait longer	5-minute timer with next-hospital alert; recommend 2 minutes for P1 [confirm]	Product owner
Internet drops in ambulances	Lost status updates	Save on the phone first; send later; hospital phones the crew if a job is not confirmed	Mobile lead
Team speed lower than planned	Go-live moves	Re-plan after Sprint 3 using real speed; cut lower-priority items first	Project manager
Late client decisions	Blocked tasks	Decision log with dates; raise in every sprint review	Project manager
Security weakness found late	Go-live delayed	Security checks in every sprint; independent test in Sprint 10	Tech lead
Key team member leaves	Work slows	Code reviews by two people; shared documentation	Tech lead

8. Quality and testing plan

QA tests every task against its acceptance criteria in ClickUp before it can be marked Done.

Test type	When	Who	Pass when
Unit and API tests	Every code change	Developers	All pass in the pipeline
Task tests (acceptance criteria)	Every task in "Ready to test"	QA	Every criterion passes on staging

Test type	When	Who	Pass when
Language tests	Every screen	QA	Works in Arabic and English, right direction
Role and scope tests	Every API	QA	No access outside scope
Network-off tests	Every mobile task	QA	Nothing lost or doubled
End-to-end case tests	Sprint 9 onward	QA	Every case path from call to discharge works
Load test	Sprint 10	QA and DevOps	Fast at 3× the busiest expected time
Security test	Sprint 10	Independent team	No high-risk findings open
User acceptance test	Sprint 11	Client with QA	Client signs off

Bug severity:

Level	Meaning	Fix by
Critical	Blocks a case, loses data, or a security hole	Same day
High	A main feature does not work, no workaround	Within the sprint
Medium	Works with a workaround	Next sprint
Low	Text, layout or small issues	When time allows

Every bug in ClickUp has: steps to repeat, expected result, actual result, screenshot or video, language and device.

9. Communication and reporting

What	Who	How often	Content
Daily stand-up	Team	Daily, 15 minutes	Progress and blockers

What	Who	How often	Content
Sprint review (demo)	Team, client, Health Authority	Every 2 weeks	Working software, accepted tasks
Sprint report	Project manager to client	Every 2 weeks	Done vs planned, next sprint goal, risks, decisions needed
Decision log	Project manager	Updated when decided	Every client decision with date and owner
Steering meeting	Client leadership and project manager	Monthly	Progress against the roadmap, big risks, scope changes

The sprint report always shows: tasks done vs planned, story points done (velocity), open bugs by severity, blocked items, and decisions the client must make.

10. Tools and where things live

What	Where
Tasks, sprints, statuses	ClickUp — Development space → **EMR-MR Sudan** folder (one list per sprint, plus "Backlog – Release 2")
Requirements	EMR-MR Sudan — Software Requirements Specification
Technical design	EMR-MR Sudan — Full System document; Technical Gaps and Go-Live Plan; Field App document
Delivery plan	EMR-MR Sudan — Delivery Plan
Designs	Figma (design system and all screens)
Code	Git repositories with the CI/CD pipeline
Servers	AWS (main and backup regions)
Presentation for management	EMR-MR Sudan — System Proposal & UI/UX Direction

11. Open decisions

- [ ] Sprint length: 2 weeks (recommended) or another length?
- [ ] Hospital answer time for P1 cases: 2 minutes (recommended) or 5 minutes like other cases?
- [ ] Which standard triage model to use?
- [ ] Who approves the patient-data rules: Health Authority, Ministry, or both?
- [ ] Written approval for storing data on AWS outside Sudan.
- [ ] Official name of the Health Authority, as it should appear in the system.
- [ ] Names of the people in each team role, to assign ClickUp tasks.